

LEVERAGE

1. Income Statement

Particulars	Amount
Sales	---
Less: Variable Cost	---
Contribution	---
Less: Fixed Cost	---
Earnings Before Interest & Tax (EBIT)	---
Less: Interest	---
Earnings Before Tax (EBT)	---
Less: Tax	---
Earnings After Tax (EAT)	---
Less: Preference Dividend	---
Earnings Available for Equity (EAFE)	---
Number of Equity Shares	---
Earnings per Share	---

2. PV Ratio = $\frac{\text{Contribution}}{\text{Sales}} \times 100 = 100 - \text{Variable Cost Ratio}$

VC Ratio = $\frac{\text{Variable Cost}}{\text{Sales}} \times 100$

3. PBT × (1 – t) = PAT

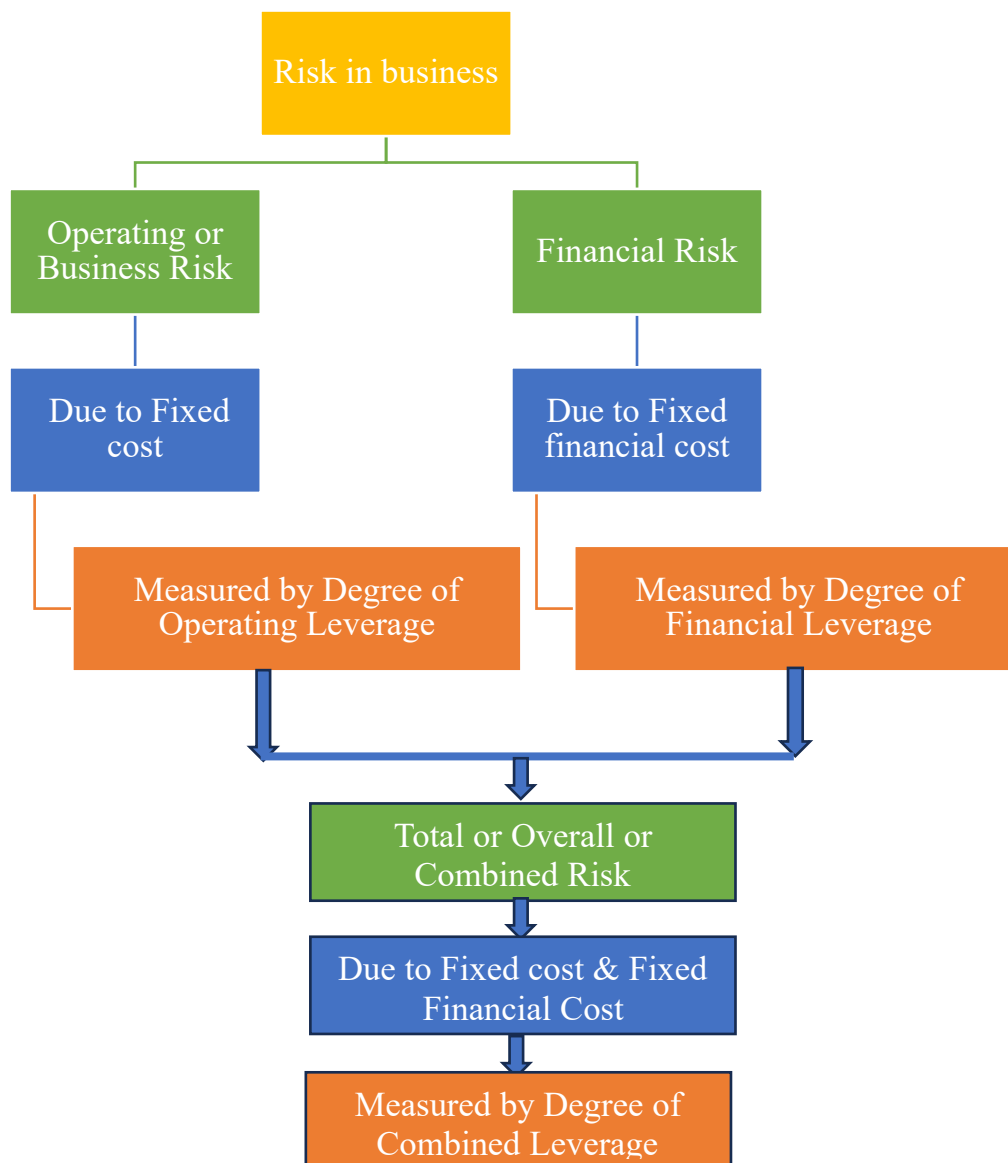
PBT = $\frac{\text{PAT}}{(1-t)}$

4. Double means increase by 100%

Triple means increase by 200%

Becomes zero means decrease by 100%

5. Risks in the business



6. Degree of Operating Leverage (DOL)

$$DOL = \frac{\% \text{ Change in EBIT}}{\% \text{ Change in Sales}}$$

$$DOL = \frac{\text{Contribution}}{EBIT}$$

$$DOL = \frac{1}{\text{Margin of Safety (MOS)}}$$

7. Degree of Financial Leverage (DFL)

$$DFL = \frac{\% \text{ Change in EPS}}{\% \text{ Change in EBIT}}$$

$$DFL \text{ (without preference shares)} = \frac{EBIT}{EBT}$$

$$DFL \text{ (with preference shares)} = \frac{EBIT}{EBT - \left(\frac{PD}{1-t}\right)}$$

8. Degree of Combined Leverage (DCL)

$$DCL = \frac{\% \text{ Change in EPS}}{\% \text{ Change in Sales}}$$

$$DCL \text{ (without preference shares)} = \frac{\text{Contribution}}{EBT}$$

$$DCL \text{ (with preference shares)} = \frac{\text{Contribution}}{EBT - \left(\frac{PD}{1-t}\right)}$$

$$DCL = DOL \times DFL$$

9. Higher the level of leverage, high will be the level of that particular risk and vice-versa.**Question – 1**

A company had the following balance sheet as on 31st March, 2021:

Liabilities	₹ in crores	Assets	₹ in crores
Equity share capital (75 lakhs shares of ₹ 10 each)	7.50	Building	12.50
Reserve and Surplus	1.50	Machinery	6.25
15% Debentures	15.00	Current Assets	
Current Liabilities	6.00	Stock	3.00
		Debtors	3.25
		Bank Balance	5.00
	30.00		30.00

The additional information given is as under:

Fixed cost per annum (excluding interest) ₹ 6 crores

Variable operating cost ratio 60%

Total assets turnover ratio 2.5

Income tax rate 40%

Calculate the following and comment:

- Earnings per share
- Operating leverage
- Financial leverage
- Combined leverage

Solution

$$\text{Total assets turnover ratio} = \frac{\text{Sales}}{\text{Total Assets}}$$

$$2.5 = \frac{\text{Sales}}{30 \text{ crores}}$$

$$\text{Sales} = ₹ 75 \text{ Crores}$$

Income Statement

<u>Particulars</u>	<u>Amount (₹)</u>
Sales	75,00,00,000
Less: Variable Cost@ 60%	<u>45,00,00,000</u>
Contribution	30,00,00,000
Less: Fixed Cost	<u>6,00,00,000</u>
EBIT	24,00,00,000
Less: Interest (15 crore × 15%)	<u>2,25,00,000</u>
EBT	21,75,00,000
Less: Income tax @ 40%	<u>8,70,00,000</u>
EAT/EAE	<u>13,05,00,000</u>

(a) Earning per share = $\frac{EAE}{\text{No. of equity shares}} = \frac{13,05,00,000}{75,00,000} = ₹ 17.40 \text{ per share}$

It indicates the amount the company earns per share. It is used as a guide for valuing the share and making investment decisions by the investor.

(b) Operating Leverage = $\frac{\text{Contribution}}{EBIT} = \frac{30,00,00,000}{24,00,00,000} = 1.25 \text{ times}$

It indicates the structure of fixed cost in the business. It indicates sensitivity of earnings before interest and tax (EBIT) to changes in sales at a particular level.

(c) Financial Leverage = $\frac{EBIT}{EBT} = \frac{24,00,00,000}{21,75,00,000} = 1.103 \text{ times}$

It indicates the use of fixed financial cost in the capital structure. It indicates sensitivity of earning per share (EPS) to changes in earnings before interest and tax (EBIT) at a particular level.

(d) Combined Leverage = $OL \times FL = 1.103 \times 1.25 = 1.379 \text{ times}$

It indicates the choice of fixed cost and fixed financial cost in the capital structure used. It indicates the sensitivity of earning per share (EPS) to changes in sales at a particular level.

Question – 2

Calculate the operating leverage, financial leverage and combined leverage from the following data under Situation I and II and Financial Plan A and B:

Installed Capacity	4,000 units
Actual Production and Sales	75% of capacity
Selling Price	₹ 30 per unit
Variable Cost	₹ 15 per unit
Fixed Cost:	
Under Situation I	₹ 15,000
Under Situation II	₹ 20,000

	Financial Plan	
	A (₹)	B (₹)
Equity	10,000	15,000
Debt (Rate of interest at 20%)	10,000	5,000
	20,000	20,000

Solution

Particulars	Situation I	Situation II
Contribution [$4,000 \times 75\% \times (30 - 15)$]	45,000	45,000
Less: fixed cost	(15,000)	(20,000)
EBIT	30,000	25,000
Operating Leverage (Contribution/EBIT)	1.5	1.8

Calculation of Financial & Combined Leverage

Financial Plan	Situation I		Situation II	
	A	B	A	B
EBIT	30,000	30,000	25,000	25,000
Less: Interest on debt	(2,000)	(1,000)	(2,000)	(1,000)
EBT	28,000	29,000	23,000	24,000
Financial Leverage (EBIT/EBT)	1.07	1.03	1.09	1.04

Question – 3

Following is the Balance Sheet of Gitashree Ltd. is given below:

Liabilities	Amount (₹)
Shareholder's Fund	
Equity Share Capital (₹ 10 each)	1,80,000
Reserve & Surplus	60,000
Non-Current Liabilities (10% Debentures)	2,40,000
Current Liabilities	1,20,000
Total	6,00,000
Non-Current Assets	4,50,000
Current Assets	1,50,000
Total	6,00,000

The company's total assets turnover ratio is 4. Its fixed operating cost is ₹ 2,00,000 and its variable operating cost ratio is 60%. The income tax rate is 30%. Calculate:

- (1) (a) Degree of operating leverage
 (b) Degree of financial leverage
 (c) Degree of combined leverage
- (2) Find out EBIT if EPS is (a) ₹ 1; (b) ₹ 2; and (c) ₹ 0.

Solution

$$\text{Total assets turnover ratio} = \frac{\text{Sales}}{\text{Total Assets}}$$

$$4 = \frac{\text{Sales}}{6,00,000}$$

$$\text{Sales} = ₹ 24,00,000$$

Income Statement

<u>Particulars</u>	<u>Amount (₹)</u>
Sales	24,00,000
Less: Variable Cost @ 60%	<u>14,40,000</u>
Contribution	9,60,000
Less: Fixed Cost	<u>2,00,000</u>
EBIT	7,60,000
Less: Interest (2,40,000 × 10%)	<u>24,000</u>
EBT	7,36,000
Less: Income tax @ 30%	<u>2,20,800</u>
EAT/EAE	<u>5,15,200</u>

$$(1) \quad (a) \quad \text{Operating Leverage} = \frac{\text{Contribution}}{\text{EBIT}} = \frac{9,60,000}{7,60,000} = 1.263 \text{ times}$$

$$(b) \quad \text{Financial Leverage} = \frac{\text{EBIT}}{\text{EBT}} = \frac{7,60,000}{7,36,000} = 1.033 \text{ times}$$

$$(c) \quad \text{Combined Leverage} = \text{OL} \times \text{FL} = 1.263 \times 1.033 = 1.304 \text{ times}$$

$$(2) \quad (a) \quad \text{EPS} = \frac{(\text{EBIT} - \text{Interest})(1-t)}{\text{No. of equity shares}}$$

$$1 = \frac{(\text{EBIT} - 24,000)(1-0.30)}{18,000}$$

$$\text{EBIT} = ₹ 49,714$$

$$(b) \quad \text{EPS} = \frac{(\text{EBIT} - \text{Interest})(1-t)}{\text{No. of equity shares}}$$

$$2 = \frac{(\text{EBIT} - 24,000)(1-0.30)}{18,000}$$

$$\text{EBIT} = ₹ 75,429$$

$$(c) \quad \text{EPS} = \frac{(\text{EBIT} - \text{Interest})(1-t)}{\text{No. of equity shares}}$$

$$0 = \frac{(\text{EBIT} - 24,000)(1-0.30)}{18,000}$$

$$\text{EBIT} = ₹ 24,000$$

Question – 4

The information related to XYZ Company Ltd. for the year ended 31st March, 2020 are as follows:

Equity Share capital of ₹ 100 each	₹ 50 lakhs
12% Bonds of ₹ 1,000 each	₹ 30 lakhs
Sales	₹ 84 lakhs
Fixed cost (excluding interest)	₹ 7.50 lakhs
Financial leverage	1.39
Profit-volume ratio	25%
Market price per equity share	₹ 200
Income tax applicable	30%

You are required to calculate:

- Operating Leverage
- Combined Leverage
- Earnings per share
- Earning Yield

Solution**Income Statement**

Particulars	Amount (₹)
Sales	84,00,000
Less: Variable cost (84,00,000 × 75%)	63,00,000
Contribution (84,00,000 × 25%)	21,00,000
Less: Fixed cost	7,50,000
EBIT	13,50,000
Less: Interest on bonds (12% × 30 lakhs)	3,60,000
Less: Other fixed interest (bal. figure)	18,777
EBT (13,50,000 ÷ 1.39)	9,71,223
Less: Tax @ 30%	2,91,367
EAT	6,79,856

$$(a) \text{ Operating Leverage} = \frac{\text{Contribution}}{\text{EBIT}} = \frac{21,00,000}{13,50,000} = 1.56 \text{ times}$$

$$(b) \text{ Combined Leverage} = \text{Operating Leverage} \times \text{Financial Leverage} = 1.56 \times 1.39 = 2.13$$

$$(c) \text{ Earnings per share (EPS)} = \frac{\text{EAT}}{\text{No. of shares outstanding}} = \frac{6,79,856}{50,000} = ₹ 13.597$$

$$(d) \text{ Earning yield} = \frac{\text{EPS}}{\text{Market price per share}} \times 100 = \frac{13.597}{200} \times 100 = 6.798\%$$

Question – 5

From the following information, prepare Income Statement of Company A & B:

Particulars	Company A	Company B
Margin of safety	0.20	0.25
Interest	₹ 3,00	₹ 2,000
Profit volume ratio	025%	33.33%
Financial Leverage	4	3
Tax rate	45%	45%

Solution**Income Statement**

Particulars	Company A (₹)	Company B (₹)
Sales	80,000	36,000
(-) Variable cost	60,000	24,000
Contribution	20,000	12,000
(-) Fixed cost	16,000	9,000
EBIT	4,000	3,000
(-) Interest	3,000	2,000
EBT	1,000	1,000
(-) Tax @ 45%	450	450
EAT	550	550

Working Notes:**(i) Company A**

$$\begin{aligned}
 \text{Financial leverage} &= \frac{EBIT}{EBIT - \text{Interest}} \\
 4 \text{ times} &= \frac{EBIT}{EBIT - 3,000} \\
 4(EBIT) - ₹ 12,000 &= EBIT \\
 EBIT &= ₹ 4,000
 \end{aligned}$$

Company B

$$\begin{aligned}
 \text{Financial leverage} &= \frac{EBIT}{EBIT - \text{Interest}} \\
 3 \text{ times} &= \frac{EBIT}{EBIT - 2,000} \\
 3(EBIT) - ₹ 6,000 &= EBIT \\
 EBIT &= ₹ 3,000
 \end{aligned}$$

(ii) Company A

$$\text{Operating leverage} = \frac{1}{\text{Margin of Safety}} = 1 \div 0.20 = 5 \text{ times}$$

$$\text{Operating leverage} = \frac{\text{contribution}}{EBIT}$$

$$5 \text{ times} = \frac{\text{contribution}}{4,000}$$

Contribution = ₹ 20,000

Company B

$$\text{Operating leverage} = \frac{1}{\text{Margin of Safety}} = 1 \div 0.25 = 4 \text{ times}$$

$$\text{Operating leverage} = \frac{\text{contribution}}{\text{EBIT}}$$

$$4 \text{ times} = \frac{\text{contribution}}{3,000}$$

Contribution = ₹ 12,000

(iii) Company A

$$\text{Sales} = \frac{\text{Contribution}}{\text{PV Ratio}} = \frac{20,000}{25\%} = ₹ 80,000$$

Company B

$$\text{Sales} = \frac{\text{Contribution}}{\text{PV Ratio}} = \frac{12,000}{33.33\%} = ₹ 36,000$$

Question – 6

The sales revenue of SK Ltd. @ ₹ 20 per unit of output is ₹ 20 lakhs and contribution is ₹ 10 lakhs. At the present level of output, the DOL of the company is 2.5. the company does not have any Preference Shares. The number of Equity Shares are 1 lakh. Applicable corporate income tax rate is 50% and the rate of interest on Debt Capital is 16% p.a. Calculate the Eps (at sales revenue of ₹ 20 lakhs) and amount of debt capital of the company if a 25% decline in Sales will wipe out EPS.

Solution

(i) Calculation of Fixed Cost

$$\text{DOL} = \frac{\text{Contribution}}{\text{Contribution-Fixed Cost}} \text{ or } 2.5 = \frac{₹10,00,000}{\text{EBIT}} \text{ or EBIT} = ₹4,00,000$$

$$\text{EBIT} = \text{Contribution} - \text{Fixed Cost}$$

$$₹4,00,000 = ₹10,00,000 - \text{Fixed Cost}$$

$$\text{Fixed Cost} = ₹10,00,000 - ₹4,00,000 = ₹6,00,000$$

(ii) Calculation of Degree of Combined Leverage (DCL)

Question says that 25% change in sales will wipe out EPS. Here, wipe out means it will reduce EPS by 100%.

$$\text{DCL} = \frac{\text{Percentage Change in EPS}}{\text{Percentage Change in Sales}} = \frac{100\%}{25\%} = 4$$

(iii) Calculation of Degree of Financial Leverage (DFL)

$$\text{DCL} = \text{DOL} \times \text{DFL}$$

$$4 = 2.5 \times \text{DFL}$$

$$\text{So, DFL} = 1.6$$

(iv) Calculation of Interest and amount of Debt

$$DFL = \frac{EBIT}{EBIT - \text{Int}} \text{ Or, } 1.6 = \frac{₹4,00,000}{₹4,00,000 - \text{Int}} \text{ Or, Int} = ₹1,50,000$$

Debt × Interest rate = Amount of Interest

$$\text{Debt} \times 16\% = ₹1,50,000$$

$$\text{Debt} = ₹9,37,500$$

(v) Calculation of Earnings per share (EPS)

$$EPS = \frac{(EBIT - \text{Int})(1 - t)}{N} = \frac{(₹4,00,000 - ₹1,50,000)0.5}{1,00,000} = ₹1.25$$

Question – 7

Information of A Ltd. is given below:

- Earnings after tax: 5% on sales
- Income tax rate: 50%
- Degree of operating leverage: 4 times
- 10% Debenture in capital structure: ₹ 3 lakhs
- Variable costs: ₹ 6 lakhs

Required:

(i) From the given data complete the following statement:

Sales	XXXX
Less: Variable costs	6,00,000
Contribution	XXXX
Less: Fixed costs	XXXX
EBIT	XXXX
Less: Interest expenses	XXXX
EBT	XXXX
Less: Income tax	XXXX
EAT	XXXX

(ii) Calculate Financial Leverage and Combined Leverage.

(iii) Calculate the percentage change in earning per share, if sales increased by 5%.

Solution

Let sales = y

$$\text{Degree of operating leverage} = \frac{\text{Contribution}}{EBIT}$$

$$4 = \frac{\text{Contribution}}{EBIT}$$

$$4(EBIT) = \text{Sales} - \text{Variable cost}$$

$$4(EBIT) = \text{Sales} - 6,00,000$$

$$EBIT = 0.25(y) - 1,50,000 \dots\dots\dots(i)$$

Also, given Earning after tax = 5% of sales

$$5\% \times \text{Sales} = (EBIT - \text{Interest})(1 - t)$$

$$0.05y = [0.25y - 1,50,000 - (3,00,000 \times 10\%)](1 - 0.50)$$

$$0.05y = (0.25y - 1,80,000)(0.50)$$

$$0.05y = 0.125y - 90,000$$

$$0.075y = 90,000$$

$$y = 12,00,000$$

$$\text{Thus, EBIT} = 0.25(12,00,000) - 1,50,000 = 1,50,000$$

$$\text{Fixed cost} = \text{Contribution} - \text{EBIT} = (12,00,000 - 6,00,000) - 1,50,000 = 4,50,000$$

Income Statement

Sales	12,00,000
Less: Variable costs	6,00,000
Contribution	6,00,000
Less: Fixed costs	4,50,000
EBIT	1,50,000
Less: Interest expenses (3,00,000×10%)	30,000
EBT	1,20,000
Less: Income tax @50%	60,000
EAT	60,000

$$(a) \text{ Financial Leverage} = \frac{EBIT}{EBT} = \frac{1,50,000}{1,20,000} = 1.25 \text{ times}$$

$$\text{Combined Leverage} = \frac{\text{Contribution}}{EBT} = \frac{6,00,000}{1,20,000} = 5 \text{ times}$$

$$(b) \text{ Combined Leverage} = \frac{\% \text{ Change in EPS}}{\% \text{ Change in Sales}}$$

$$5 = \frac{\% \text{ Change in EPS}}{+5}$$

$$\% \text{ change in EPS} = +25\%$$

Thus, EPS increases by 25.

Question – 8

From the following information for financial year 2023-24,

Financial Leverage (FL) = 4

P/V Ratio = 40%

Tax rate = 30%

Depreciation (part of manufacturing overheads) = ₹10,000

Preference dividend is 15% of Operating Profit

Cash Breakeven Sales = ₹2,25,000

Equity Share Capital = ₹1,00,000

Reserves & Surplus as on 31.03.2023 = ₹35637

Particulars	Amount (₹)
Sales	???
(-) Variable Cost	???
Contribution	???
(-) Fixed cost	???
EBIT	???
(-) Interest expense	57,400

EBT	???
(-) Tax	???
EAT	???
(-) Preference Dividend	???
Earnings for equity shareholders	???

CALCULATE –

- Complete the Income statement for FY 2023-24
- Operating Leverage & Combined Leverage
- Percentage change in EPS, if sales increase and decreases by 7%
- Calculate Return on Equity Shareholders funds on 31.03.24
- Amount of Debt, if post tax interest rate is 6.65%

Solution

WN1 – Calculation of EBIT using Financial Leverage

$$FL = \frac{EBIT}{(EBIT - \text{Int.}) - \frac{D_p}{1-t}}$$

Let EBIT be 'X'

$$\text{Preference Dividend (Dp)} = 0.15X$$

$$4 = \frac{X}{(X - 57,400) - \frac{0.15X}{1-0.3}}$$

$$3.16X - 2,29,600 = X$$

$$X = \text{EBIT} = ₹ 1,06,296$$

$$\text{Preference Dividend} = 1,06,296 \times 0.15 = ₹ 15,944$$

WN2 – Calculation of Total Fixed Cost

$$\begin{aligned} \text{Cash Fixed Cost} &= \text{Cash Breakeven sales} \times \text{PV Ratio} \\ &= 2,25,000 \times 0.40 \\ &= ₹ 90,000 \end{aligned}$$

$$\begin{aligned} \text{Total Fixed Cost} &= 90,000 + 10,000 \text{ (Depreciation)} \\ &= ₹ 1,00,000 \end{aligned}$$

$$\text{Therefore Contribution} = 1,06,296 + 1,00,000 = 2,06,296$$

i. Income Statement for FY 2023-24:

Particulars	Amount (₹)
Sales (2,06,296 / 0.40)	5,15,740
(-) Variable Cost	(3,09,444)
Contribution	2,06,296

(-) Fixed Cost (See Wn 2)	(1,00,000)
EBIT (See Wn 1)	1,06,296
(-) Interest Exp	(57,400)
EBT	48,896
(-) Tax	(14,669)
EAT	34,227
(-) Preference Dividend (1,06,296 X 15%)	(15,944)
Earnings For Equity Shareholders	18,283

ii. **Operating Leverage** = $\frac{\text{Contribution}}{\text{EBIT}}$

$$= 2,06,296 / 1,06,296$$

$$= \mathbf{1.94 \text{ times}}$$

Combined Leverage = OL x FL

$$= 4 \times 1.94$$

$$= \mathbf{7.76 \text{ times}}$$

- iii. **Combined leverage is the study from Sales to Eps, where combined leverage measures the % change in EPS due to increase/decrease in % change in sales**

$$\text{CL} = \% \text{ Change in EPS} / \% \text{ Change in sales}$$

$$7.76 = \% \text{ Change in EPS} / 7$$

$$\text{Therefore, \% Change in EPS} = 54.32\%$$

If sales increase by 7%, then EPS will rise by 54.32% and on the contrary if sales decrease by 7%, then EPS will fall by 54.32%.

- iv. **Return on Equity shareholders' funds**

$$= \frac{\text{Earnings for Equity holders}}{\text{Total Eq. Shareholders' funds}}$$

$$= 18,283 / 1,53,920$$

$$= \mathbf{11.88\%}$$

- v. **Calculation of Amount of Debt**

$$\text{Pre-tax Interest \%} = \frac{\text{Interest rate (post tax)}}{1 - t}$$

$$= 6.65 / 0.70$$

$$= 9.5\%$$

$$\text{Debt Amount} = \frac{\text{Interest (Rs.)}}{\text{Interest(\%)}}$$

$$= 57,400 / 0.095$$

$$= \mathbf{₹ 6,04,210}$$

10. Relation between BEP, FC and DOL

<u>DOL</u>	<u>Fixed Cost</u>	<u>Operating BEP</u>	<u>MOS</u>
High	High	High	Low
Low	Low	Low	High

11. DOL Analysis

Situation	Result or Interpretation
No Fixed Cost	- DOL = 1 - No operating risk
High Fixed Cost	- High BEP - High DOL
Low Fixed Cost	- Low BEP - Low DOL
Sales > Operating BEP i.e. EBIT > 0	- Existing Profit - DOL is in favour
Sales < Operating BEP i.e. EBIT < 0	- Existing Loss - DOL is against

12. Financial BEP

Level of EBIT at which EPS is zero

$$\text{Financial BEP} = \text{Interest} + \frac{PD}{(1-t)}$$

$$\text{EPS} = \frac{(EBIT - \text{Interest})(1-t) - \text{Pref. Dividend}}{\text{No. of Equity Shares}}$$

13. Analysis of DFL

Situation	Result or Interpretation
No Fixed Finance Cost	- DFL = 1 - No financial risk
High Fixed Financial Cost	- High Financial BEP - High DFL

Low Fixed Financial Cost	• Low Financial BEP • Low DFL
EBIT > Financial BEP i.e. EPS > 0	• EPS is positive • DFL is in favour
EBIT < Financial BEP i.e. EPS < 0	• EPS is negative • DFL is against

14. Financial Leverage as a Double Edge Sword

For Debt:

- If Rate of Interest < Return on Investment (ROI) Favourable
 If Rate of Interest > Return on Investment (ROI) Unfavourable
 If Rate of Interest = Return on Investment (ROI) Indifferent

For Preference:

- If Rate of Preference Dividend < ROI(1 - t) Favourable
 If Rate of Preference Dividend > ROI(1 - t) Unfavourable
 If Rate of Preference Dividend = ROI(1 - t) Indifferent

15. Trading on Equity

When the quantity of fixed cost fund is relatively high in comparison to equity capital it is said that the firm is “trading on equity”.

16. Analysis of DCL

DOL	DFL	Result or Interpretation
High	High	• High risky situation
Low	Low	• Low risk situation
High	Low	• Moderate risk • EBIT is low • No benefit of trading on equity
Low	High	• Moderate risk • EBIT is high • Benefit of trading on equity available

Question – 9

Details of a company for the year ended 31st March, 2022 are given below:

Sales	₹ 86 lakhs
Profit Volume (P/V) Ratio	35%
Fixed cost excluding interest expenses	₹ 10 lakhs
10% Debt	₹ 55 lakhs
Equity Share Capital of ₹ 10 each	₹ 75 lakhs
Income Tax rate	40%

Required:

- Determine company's return on capital employed (pre-tax) and Eps.
- Does the company have a favourable financial leverage?
- Calculate operating, financial and combine leverages of the company
- Calculate percentage change in EBIT, if sales increases by 10%.
- At what level of sales, the Earning before Tax (EBT) of the company will be equal to zero?
- Calculate percentage change in EBT, if EBIT increases by 10%.

Solution**Income Statement**

Particulars	Amount (₹)
Sales	86,00,000
Less: Variable cost (86,00,000 × 65%)	55,90,000
Contribution	30,10,000
Less: Fixed cost	10,00,000
EBIT	20,10,000
Less: Interest (10% × 55,00,000)	5,50,000
EBT	14,60,000
Less: Tax @ 40%	5,84,000
EAT/EAE	8,76,000

$$(i) \quad \text{Return on capital employed} = \frac{EBIT}{\text{Capital employed}} \times 100 = \frac{20,10,000}{1,30,00,000} \times 100 = 15.46\%$$

$$\text{Earning per share} = \frac{EAE}{\text{No. of Equity Shares}} = \frac{8,76,000}{7,50,000} = ₹ 1.168$$

- (ii) Since, the return on capital employed (15.46%) is more than the interest rate (10%), thus the company has a favourable financial leverage.

$$(iii) \quad \text{Operating leverage} = \frac{\text{Contribution}}{EBIT} = \frac{30,10,000}{20,10,000} = 1.498 \text{ times}$$

$$\text{Financial leverage} = \frac{EBIT}{EBT} = \frac{20,10,000}{14,60,000} = 1.377 \text{ times}$$

$$\text{Combined leverage} = \frac{\text{Contribution}}{EBT} = \frac{30,10,000}{14,60,000} = 2.062 \text{ times}$$

$$(iv) \quad \text{Operating leverage} = \frac{\% \text{ Change in EBIT}}{\% \text{ Change in Sales}}$$

$$1.498 = \frac{\% \text{ Change in EBIT}}{+10}$$

$$\% \text{ Change in EBIT} = +14.98$$

Thus, EBIT increases by 14.98%

$$(v) \quad \text{Required sales} = \frac{\text{Fixed cost} + \text{Interest}}{\text{PV Ratio}} = \frac{(10,00,000 + 5,50,000)}{35\%} = ₹ 44,28,571$$

$$(vi) \quad \text{Financial leverage} = \frac{\% \text{ Change in EBT}}{\% \text{ Change in EBIT}}$$

$$1.377 = \frac{\% \text{ Change in EBT}}{+10}$$

$$\% \text{ change in EBT} = 13.77\%$$